

WASP-50b

R-0754

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-50_231104 · created 2026-07-28T03:12:57+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2460252.852 +/- 0.099 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.168 +/- 0.051
Transit depth (Rp/Rs)^2	2.83 +/- 1.7 [%]
Orbital Inclination (inc)	89.73 +/- 2.85
Airmass coefficient 1 (a1)	46.65 +/- 33.0
Airmass coefficient 2 (a2)	-2.4 +/- 1.64
Scatter in the residuals of the lightcurve fit is	102.86 %
Best Comparison Star	None
Optimal Aperture	0.75
Optimal Annulus	3.0
Transit Duration (day)	0.044 +/- 0.037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.168 ± 0.051 vs 0.1404 (deviation 0.53σ)
Duration fit vs published	0.044 d vs 0.0754 d (deviation 0.83σ)
Mid-time O–C	88.2 min
Depth SNR	3.2
Residual scatter	102.86 %

Light curve

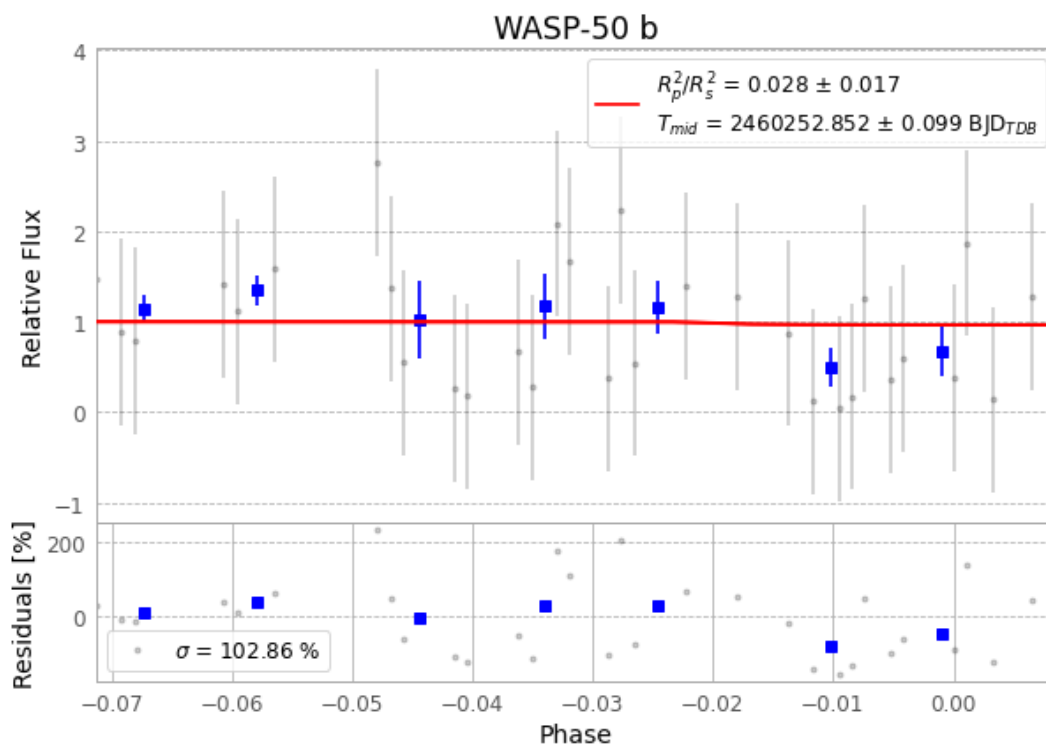


Figure 1 — FinalLightCurve_WASP-50 b_2023-11-03.png (SHA-256 ce483c80caf7f61f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [328, 400], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 ad6f36f784b0bdf4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

77 FITS frames (51 MB), WASP-50231104045715.FITS → WASP-50231104084516.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-50-231104.log (d3874292ccdc...), FinalLightCurve_WASP-50 b_2023-11-03.png (ce483c80caf7...), FinalLightCurve_WASP-50 b_2023-11-03.csv (14203d31379a...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 03:12Z.